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## Assignment Cover Page

|                   |                  |                     |                                  |
|-------------------|------------------|---------------------|----------------------------------|
| Assignment Title: | Final-Assignment |                     |                                  |
| Assignment No:    | 01               | Date of Submission: | 10 December 2021                 |
| Course Title:     | Compiler Design  |                     |                                  |
| Course Code:      | CSC3216          | Section:            | G                                |
| Semester:         | Fall             | 2021-22             | Course Teacher: Md. Masum Billah |

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| 1  | Saidul Mursalin Khan | 19-41766-3 | BSc [CSE]       |           |
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|                  |                |  |
|                  |                |  |



1. Check the grammar below and justify whether the grammar is LL(1) or not:

$S \rightarrow U$

$U \rightarrow T a U$

$U \rightarrow T a T$

$T \rightarrow a T b T$

$T \rightarrow b T a T$

$T \rightarrow d$

2. Which of the following grammars are ambiguous? Show **LL(1) parsing table** using **First** and **Follow** to determine whether the grammar is ambiguous or not.

1)  $S \rightarrow 0A \mid 1B$

$A \rightarrow 0AA \mid 1S \mid 1$

$B \rightarrow 1BB \mid 0S \mid 0$

2)  $S \rightarrow aSb \mid bSa \mid SS \mid \epsilon$

3)  $S \rightarrow 1S1 \mid T$

$T \rightarrow 1X1 \mid X$

$X \rightarrow 0X0 \mid 1$

3. Design a context free grammar by your own. Construct a **LL(1) parsing table** using **First** and **Follow** and show **stack movement** for any one input string.

### Instructions For Preparing Your Assignment:

- Your assignment must be in the form of **handwritten hard copy**, not the typing copy.
- You can use a **scanner to create a PDF file** of your completed handwritten assignment, or you can **use your smartphone to take images of your work that can be compiled in a Word document and convert it into PDF**. Please check the video tutorial who cannot understand how to make a PDF.
  - ~ Scanner: (<https://www.youtube.com/watch?v=fPp8Ox7qfCM>)
  - ~ Images: (<https://www.youtube.com/watch?v=5K7xxQ8np90>)
- Rename the PDF as your student ID\_CD\_Asg\_Final\_Sec **(Example: 19-12345-2\_CD\_Asg\_Final\_Term.pdf)**
- **Copy and paste is prohibited.** I will check everyone's assignment carefully.
- Submit the assignment on 10 December 2021 (Friday) 09:59 PM. **Deadline will be strictly followed** and marks will be deducted who misses submission deadline.
- You must **add a cover page**. The key contents of your cover page should be (Course Title, Submission Type [e.g. Final-Assignment], Your Name, ID, Section, Department and Date of Submission).

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Ans: to: the: Q: NO: -1

For justifying whether a grammar LL1 or not, we need to do the first and follow and also we need to eliminate left recursion if it is there.

Given grammar:

$$S \rightarrow U$$

$$U \rightarrow TaU$$

$$U \rightarrow TaT$$

$$T \rightarrow aTbT$$

$$T \rightarrow bTaT$$

$$T \rightarrow d$$

Which can be re-written as

$$S \rightarrow U$$

$$U \rightarrow TaU \mid TaT$$

$$T \rightarrow aTbT \mid bTaT \mid d$$

There is no left recursion so there is no need of left factoring.

Now we will generate the first and follow for the given grammar.

First Set:

$$S = \{ \text{First}(U) \} = \{ a, b, d \} \quad U \leftarrow Z$$

$$U = \{ a, b, d \}$$

$$T = \{ a, b, d \}$$

Follow Set:

$$S = \{ \$ \}$$

$$U = \{ \text{Follow}(S) \} = \{ \$ \}$$

$$\begin{aligned} T &= \{ \text{First}(a) \cup \text{First}(a) \cup \text{Follow}(U) \cup \text{First}(b) \\ &\quad \cup \text{First}(a) \cup \text{Follow}(T) \} \\ &= \{ a, \$, b \} \end{aligned}$$

Now generating the parsing table from First and Follow Set:

|   | a                                          | b                                          | d                                          | \$ |
|---|--------------------------------------------|--------------------------------------------|--------------------------------------------|----|
| S | $S \rightarrow U$                          | $S \rightarrow U$                          | $S \rightarrow U$                          |    |
| U | $U \rightarrow TaU$<br>$U \rightarrow TaT$ | $U \rightarrow TaT$<br>$U \rightarrow TaU$ | $U \rightarrow TaT$<br>$U \rightarrow TaU$ |    |
| T | $T \rightarrow aTbT$                       | $T \rightarrow bTaT$                       | $T \rightarrow d$                          |    |

As we can generate first and follow set and also the parsing table we can say that the grammar is ~~not~~ not LL1. Because there is multiple data in cells. The grammar is ambiguous.

Ans: To: The: Q: No: - 2

1)  $S \rightarrow 0A \mid 1B$

$A \rightarrow 0AA \mid 1S \mid 1$

$B \rightarrow 1BB \mid 0S \mid 0$

First Set :

$$S = \{0, 1\}$$

$$A = \{0, 1\}$$

$$B = \{1, 0\}$$

Follow Set:

$$S = \{ \$ \cup \text{First}(A) \cup \text{Follow}(A) \cup \text{Follow}(B) \}$$

$$= \{ \$, 0, 1 \}$$

$$A = \{ \text{Follow}(S) \cup \text{First}(A) \cup \text{Follow}(A) \}$$

$$= \{ \$, 0, 1 \}$$

$$B = \{ \text{Follow}(S) \cup \text{First}(B) \cup \text{Follow}(B) \}$$

$$= \{ \$, 0, 1 \}$$

## Parsing Table:

|   | 0                                       | 1                                       | \$ |
|---|-----------------------------------------|-----------------------------------------|----|
| S | $S \rightarrow 0A$                      | $S \rightarrow 1B$                      |    |
| A | $A \rightarrow 0AA$                     | $A \rightarrow 1S$<br>$A \rightarrow 1$ |    |
| B | $B \rightarrow 0S$<br>$B \rightarrow 0$ | $B \rightarrow 1BB$                     |    |

As there is two cells with multiple data as we know by the rules,

(a) the grammar is ambiguous.

$$\{1, 0, \$\} =$$

$$\{(A) \text{ follow } U (A) \text{ first } U (2) \text{ follow}\} = A$$

$$\{1, 0, \$\} =$$

$$\{(B) \text{ follow } U (B) \text{ first } U (2) \text{ follow}\} = B$$

$$\{1, 0, \$\} =$$

$$2) S \rightarrow asb \mid bsa \mid ss \mid \epsilon$$

The following grammar has left recursion in one of its rules  $S \rightarrow ss$ .

We need to eliminate it first.

$$\underline{S} \rightarrow \underline{S} \underline{S} \mid \underline{\epsilon}$$

$$A' \quad A \quad \alpha \quad \beta$$

$$S' \rightarrow \epsilon \mid SS' \{a, b, \$\}$$

So after eliminating left recursion

$$S \rightarrow asb \mid bsa \mid S' \mid \epsilon$$

$$S' \rightarrow \epsilon \mid SS'$$



Now,

First set:

$$S = \{a, b, \text{First}(s'), \epsilon\}$$

$$= \{a, b, \epsilon\}$$

$$s' = \{\epsilon, a, b\}$$

Follow Set:

$$S = \{\$ \cup \text{First}\{b\} \cup \text{First}\{a\}\}$$

$$= \{\$, b, a\}$$

$$s' = \{\text{Follow}(s)\}$$

$$= \{\$, b, a\}$$

Now generating the parsing table,

|    | a                                                 | b                                                 | \$                        |
|----|---------------------------------------------------|---------------------------------------------------|---------------------------|
| S  | $S \rightarrow aSb$<br>$S \rightarrow \epsilon$   | $S \rightarrow bSa$<br>$S \rightarrow \epsilon$   | $S \rightarrow \epsilon$  |
| S' | $S' \rightarrow \epsilon$<br>$S' \rightarrow SS'$ | $S' \rightarrow \epsilon$<br>$S' \rightarrow SS'$ | $S' \rightarrow \epsilon$ |

As there is 4 cells with multiple data as by the rules we can say the grammar is ambiguous.

$$\{1, 1\} = \{(1) \text{ Follow } V, 1\} = 2$$

$$\{1, 1\} = \{(2) \text{ Follow } V\} = 1$$

$$3) S \rightarrow |S| | T$$

$$T \rightarrow |X| | x$$

$$x \rightarrow 0x0, 1$$

First set:

$$S = \{1 \cup \text{First}\{T\}\} = \{1, 0\}$$

$$T = \{1 \cup \text{First}\{x\}\} = \{1, 0\}$$

$$x = \{0, 1\}$$

Follow set:

$$S = \{\$ \cup \text{First}(1)\} = \{\$, 1\}$$

$$T = \{\text{Follow}(S)\} = \{\$, 1\}$$

$$\begin{aligned} x &= \{\text{First}(1) \cup \text{Follow}(T) \cup \text{First}(0)\} \\ &= \{\$, 1, 0\} \end{aligned}$$

## Generating the parsing table

|   | 0                   | 1                   | \$ |
|---|---------------------|---------------------|----|
| S | $S \rightarrow T$   | $S \rightarrow 1S1$ |    |
| T | $T \rightarrow X$   | $T \rightarrow 1X1$ |    |
| X | $X \rightarrow 0X0$ | $X \rightarrow 1$   |    |

From the parsing table we can see there is no cell with multiple rules. So, as of we know, this grammar is not ambiguous.

Ans: To: The: Q: No: - 3

(First designing own CFG.)

$$A \rightarrow PaT | Rc | \epsilon$$

$$R \rightarrow PrI | OtY$$

$$P \rightarrow AbR | \epsilon$$

$$I \rightarrow Ad$$

First Set:

$$A = \{a, c, \epsilon\}$$

$$R = \{r, t, y\}$$

$$P = \{b, \epsilon\}$$

$$I = \{d\}$$

Follow Set:

$$A = \{ \$ \cup \text{First}\{b\} \cup \text{First}\{d\} \}$$

$$= \{ \$, b, d \} \quad T \rightarrow A$$

$$P = \{ \text{First}\{a\} \cup \text{First}\{r\} \}$$

$$= \{ a, r \} \quad A \rightarrow P$$

$$R = \{ c \cup \text{Follow}(P) \}$$

$$= \{ a, r, c \}$$

$$I = \{ \text{Follow}(R) \}$$

$$= \{ a, r, c \}$$

Generating Parsing Table

|   | a                   | c                  | r                   | t                  | b                   | d                  | \$                |
|---|---------------------|--------------------|---------------------|--------------------|---------------------|--------------------|-------------------|
| A | $A \rightarrow PaT$ | $A \rightarrow Rc$ |                     |                    | $A \rightarrow E$   | $A \rightarrow E$  | $A \rightarrow E$ |
| R |                     |                    | $R \rightarrow PrI$ | $R \rightarrow Ot$ |                     |                    |                   |
| P | $P \rightarrow E$   |                    | $P \rightarrow E$   |                    | $P \rightarrow AbR$ |                    |                   |
| I |                     |                    |                     |                    |                     | $I \rightarrow Ad$ |                   |

## Stack Movement

Input String: abty

| Trace |        |                          | Tree |
|-------|--------|--------------------------|------|
| Stack | input  | Rule                     |      |
| \$A   | abty\$ |                          |      |
| \$Tap | abty\$ | $A \rightarrow PaT$      |      |
| \$Ia  | abty\$ | $P \rightarrow \epsilon$ |      |
| \$T   | bty\$  | terminal found           |      |

So, the following string cannot be generated by this grammar.